Teacher Wu Kuan | Real Exam #2 — Taiwan GSAT Math B (2026) Q14: Adjacent Arrangements

English lecture notes · YouTube Short EYMRTbRv71k

1. The trap question

Arrange 1–7 so that 3 & 4, 5 & 6, and 6 & 7 are each adjacent. How many ways?

2. The method

• Glue adjacent items into blocks; arrange the blocks; then arrange inside each block.
• Trap: 6 is in two conditions, so 5-6-7 is ONE block with 6 in the middle: 567 or 765 (2 ways).
• Units [34], [567], 1, 2 → 4! = 24; × 2 × 2 = 96.

3. Parent check method

Have your child write out the blocks as boxes and list the internal orders before multiplying.

Practice

1. Arrange 1, 2, 3, 4, 5 in a row with 1 and 2 adjacent. How many ways?
2. Arrange 1–6 with 1, 2 adjacent and 3, 4 adjacent.
3. Arrange 1–5 with 1, 2 adjacent and 2, 3 adjacent.
4. Arrange 1–5 with 1, 2 adjacent but 2, 3 NOT adjacent.
5. Arrange 1–7 with 1-2, 2-3, 3-4 each adjacent, and 5, 6 NOT adjacent.

Answers

1. 48
2. 96
3. 12
4. 36
5. 24

Saber Math · Teacher Wu Kuan (YouTube: Saber Math) · Free first-week trial of Kuan Weekly practice (email sign-up): https://kuan-weekly.vercel.app/sample · More: https://kuanmath.vercel.app · SAT notes: DM IG @saber_math